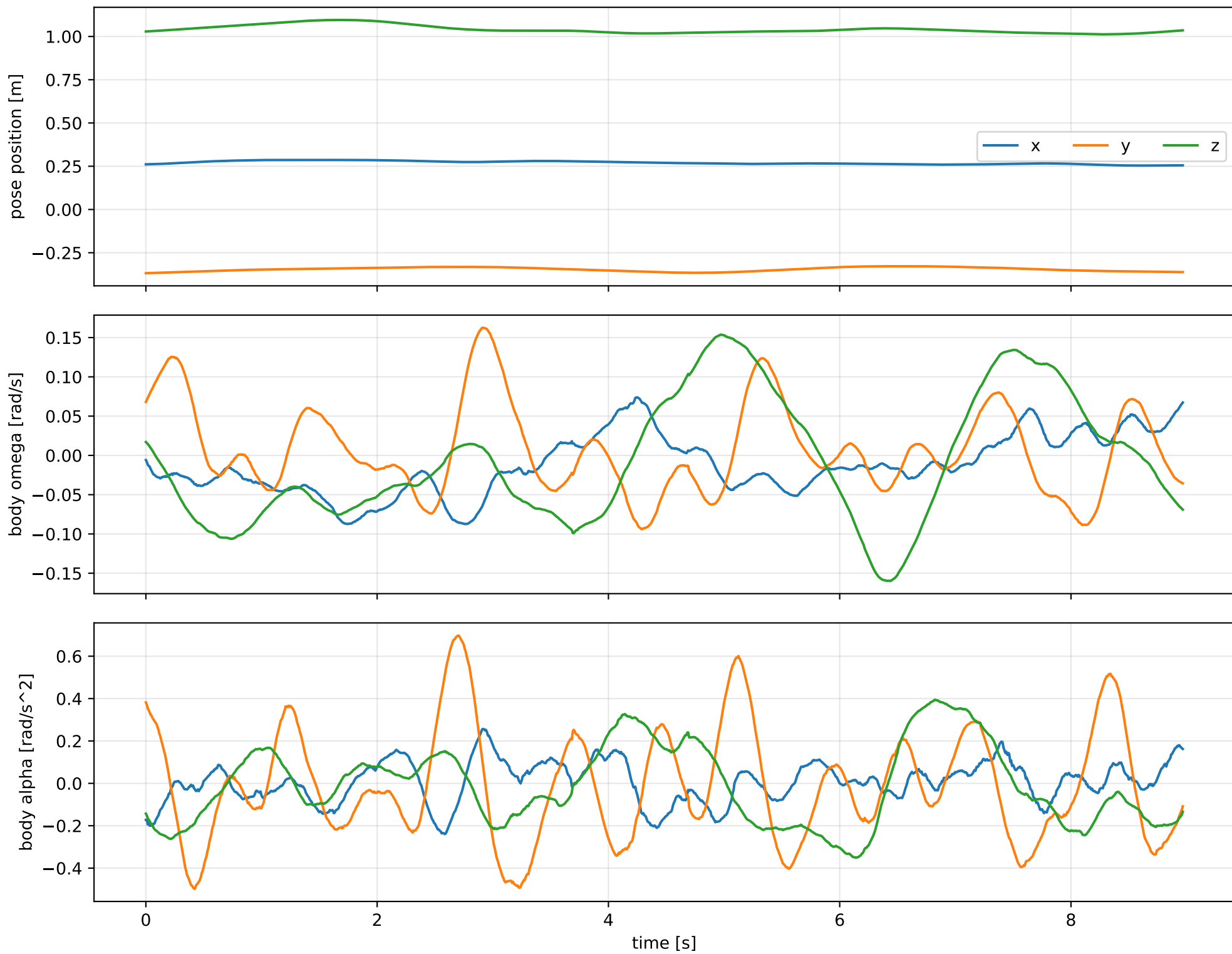
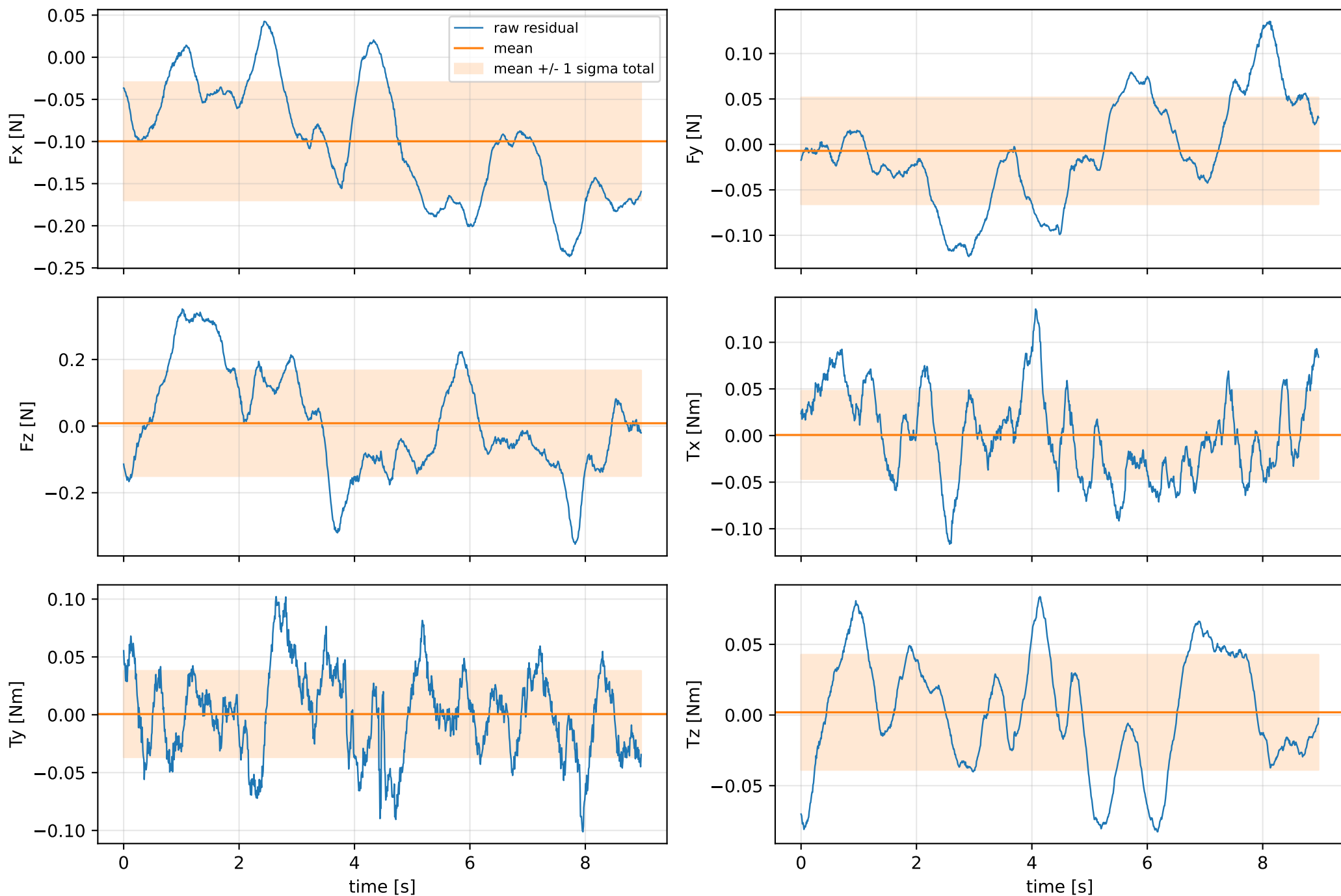


inertia_over_mass_xy_nominal: SG trajectory and derived motion



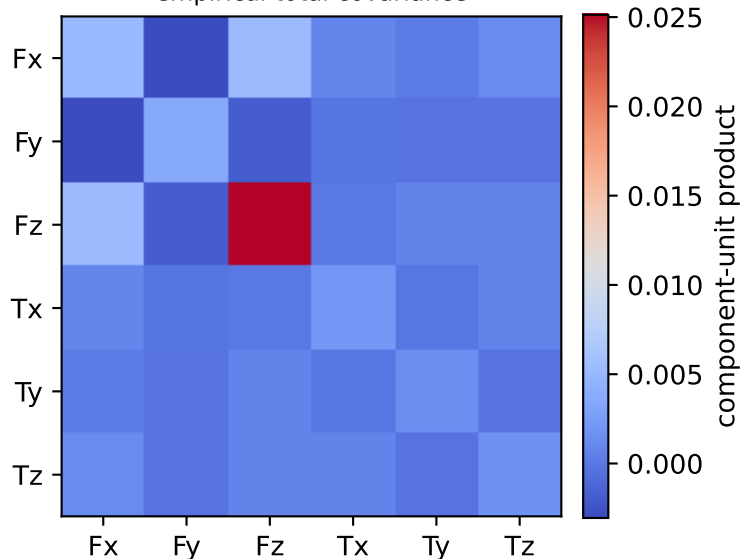
inertia_over_mass_xy_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



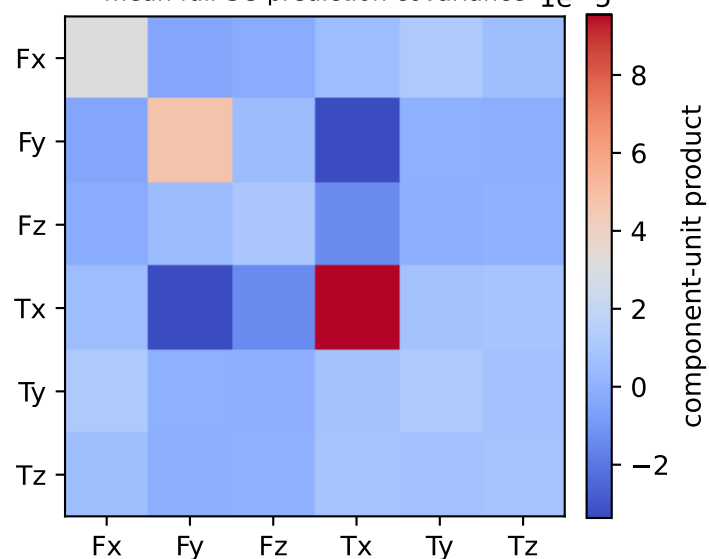
vector RMS about zero: force 0.20865 N, torque 0.07268 Nm; component std about mean: force [0.0701 0.0587 0.1585], torque [0.0472 0.0373 0.0407]

inertia_over_mass_xy_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.002 0.003 0.006 0.027]

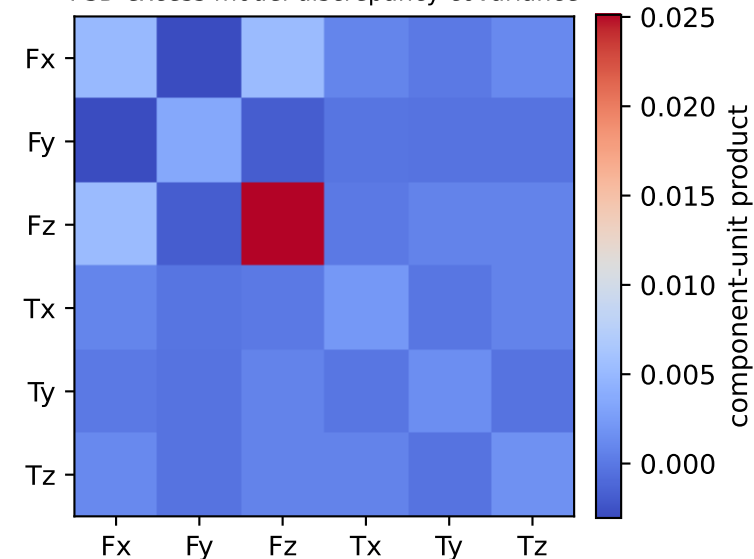
empirical total covariance



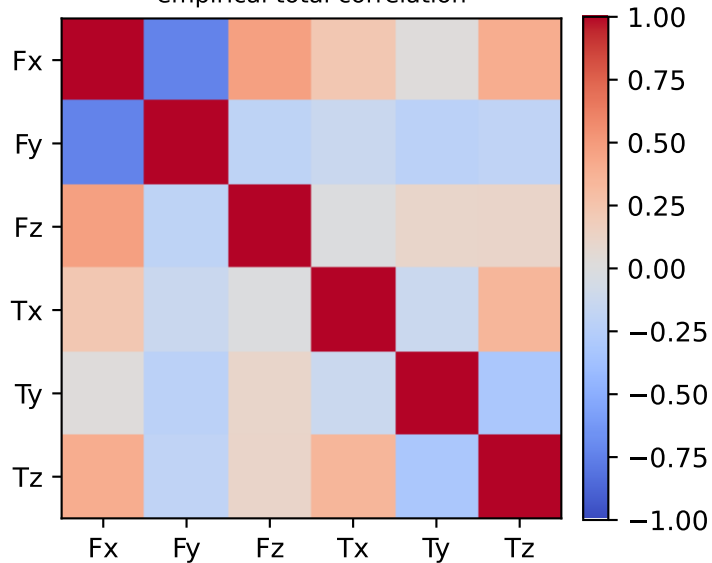
mean full-SG prediction covariance $1e-5$



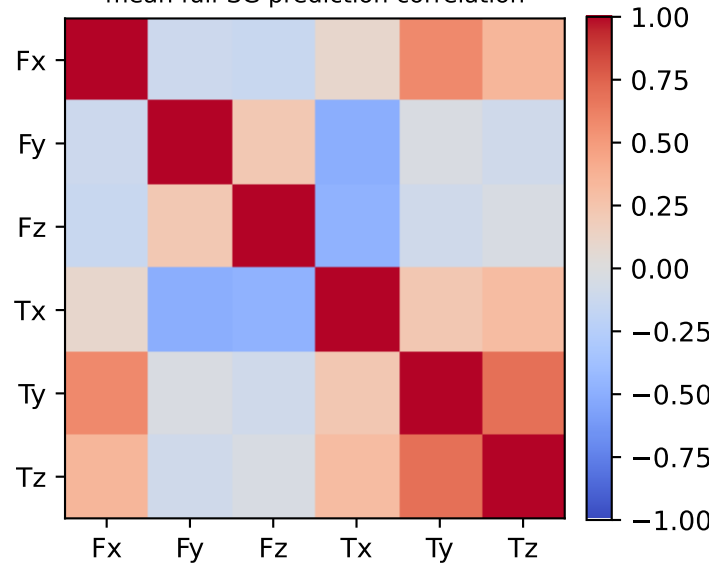
PSD excess model discrepancy covariance



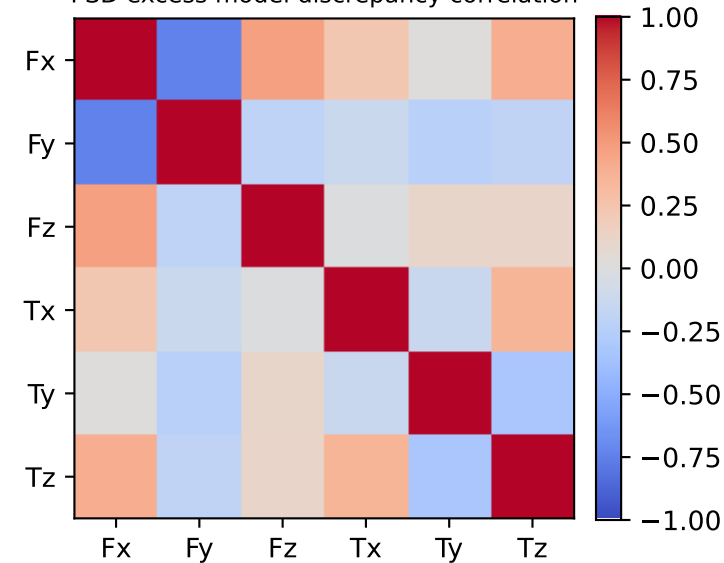
empirical total correlation



mean full-SG prediction correlation

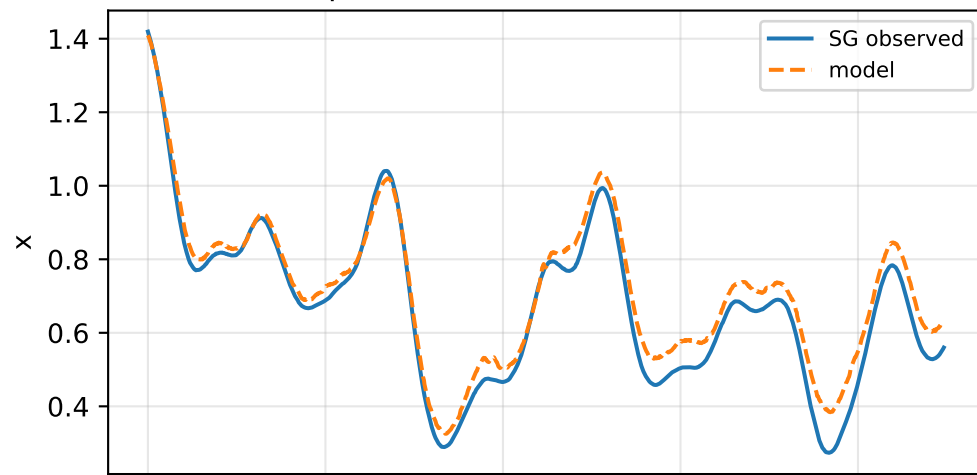


PSD excess model discrepancy correlation

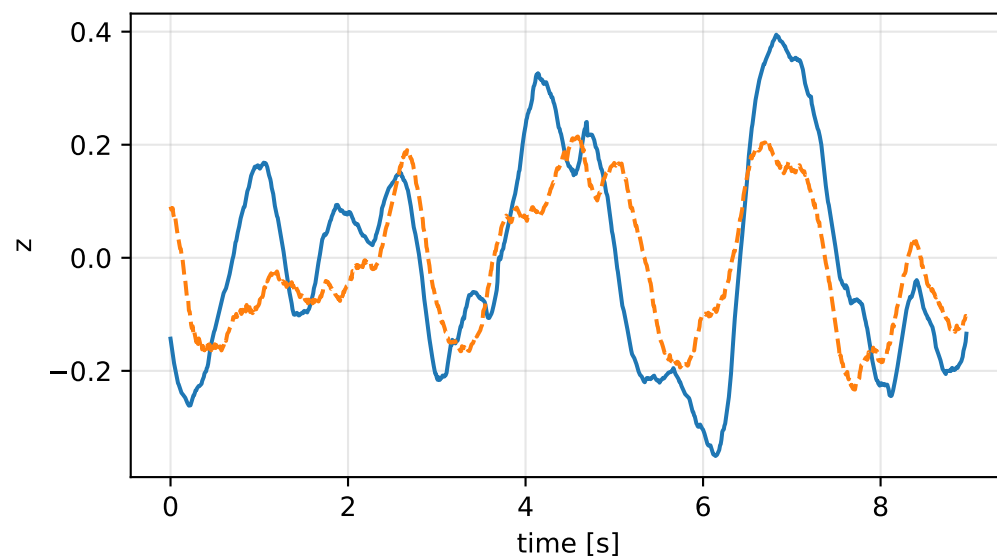
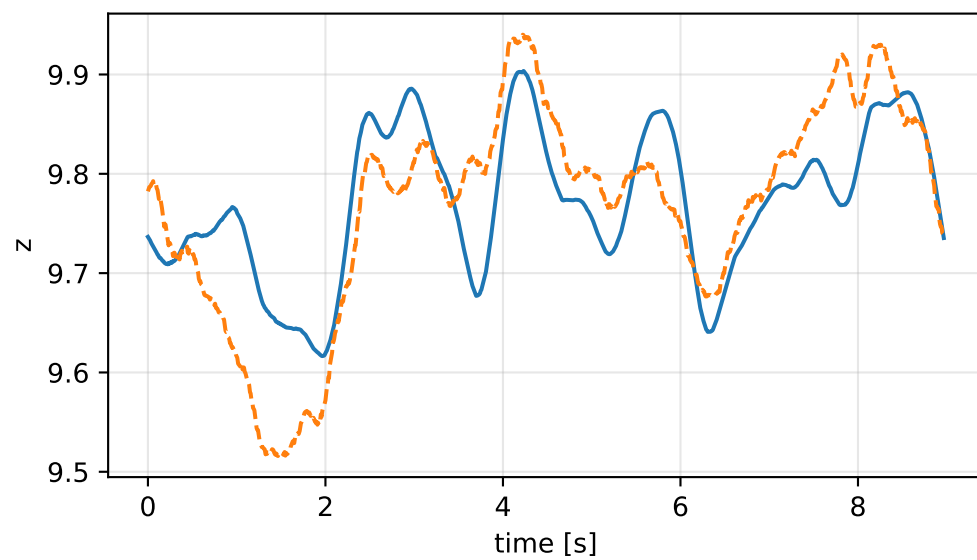
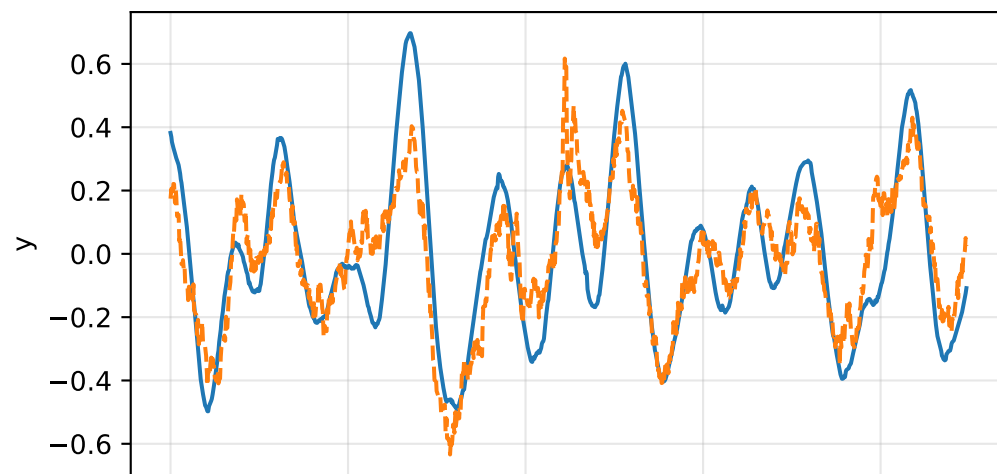
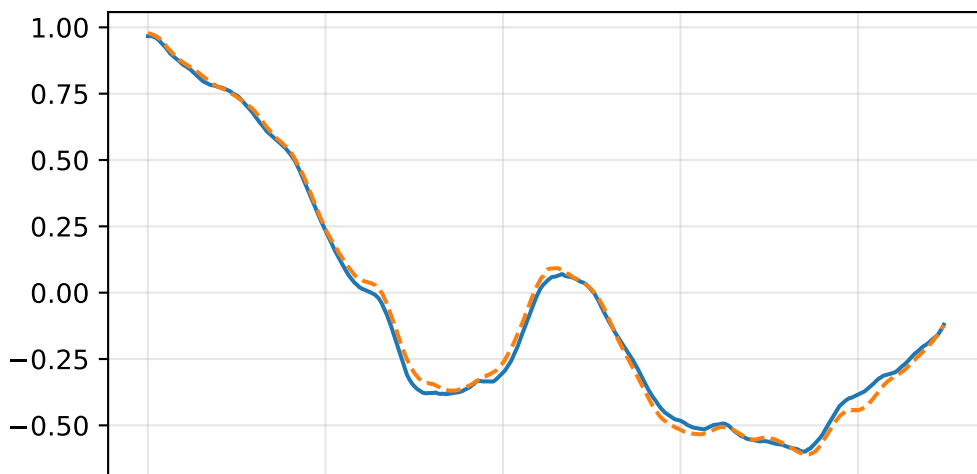
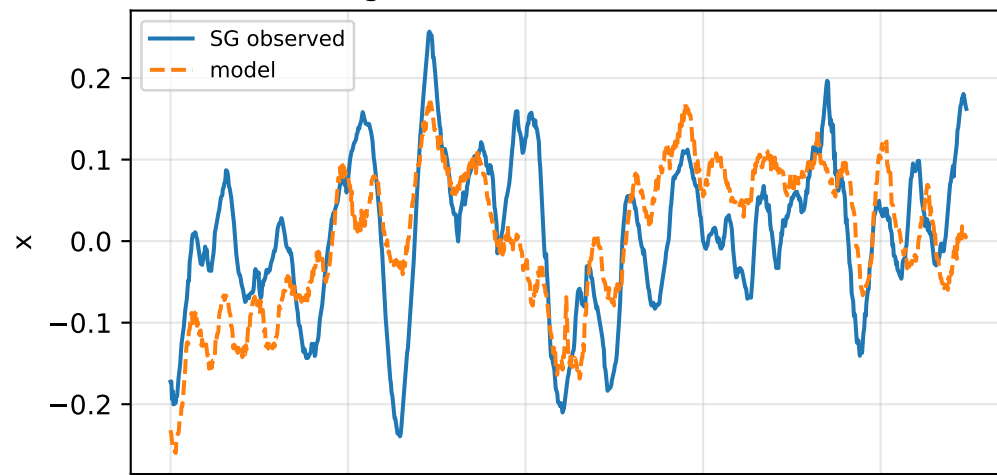


inertia_over_mass_xy_nominal: acceleration residual objective

specific acceleration [m/s²]

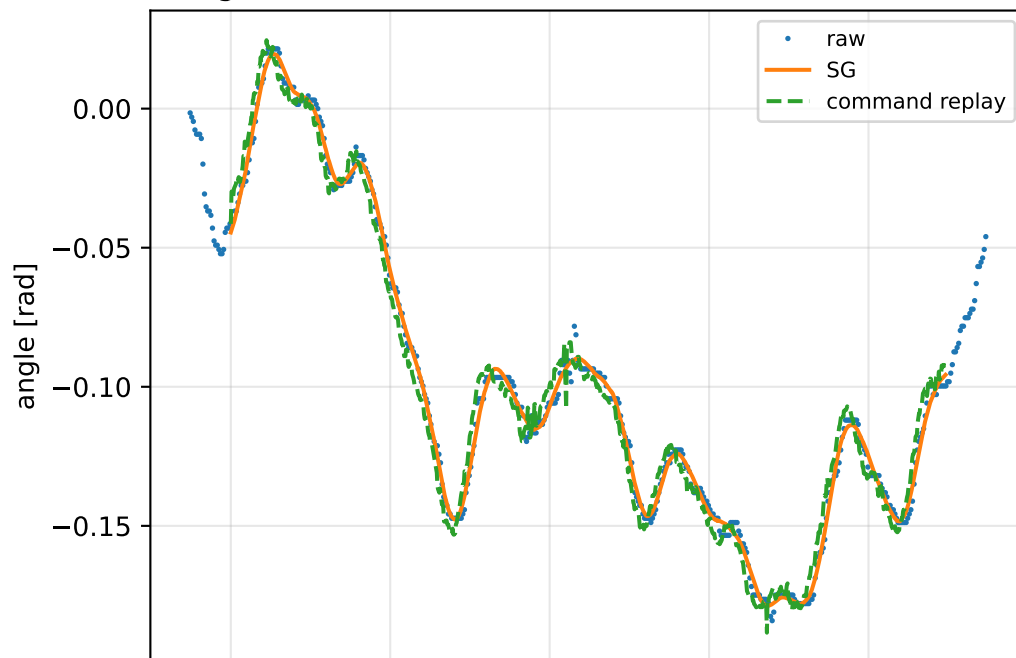


angular acceleration [rad/s²]

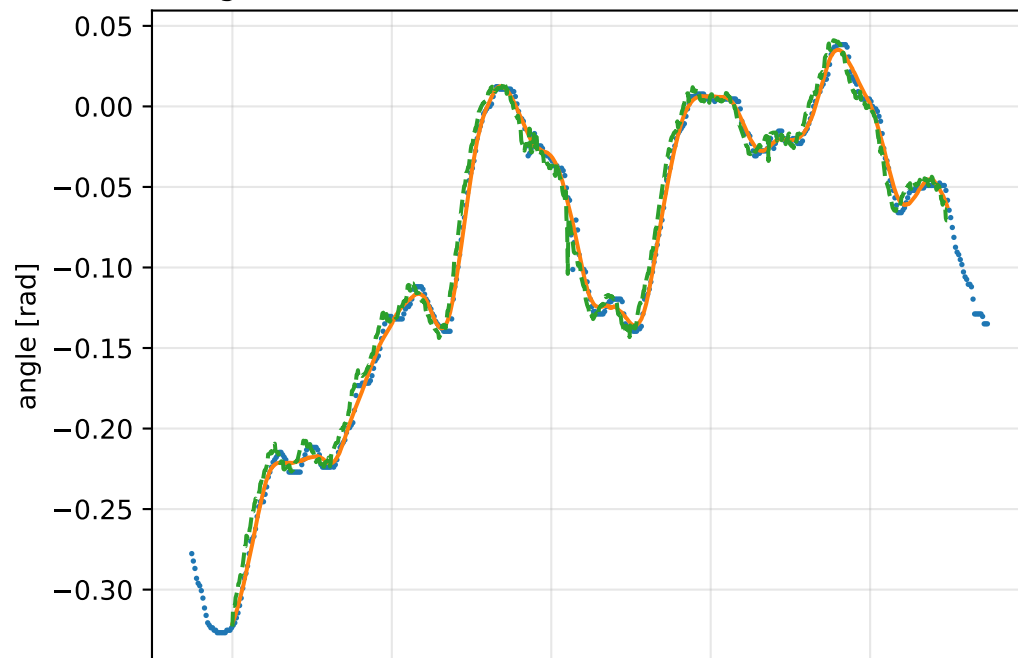


inertia_over_mass_xy_nominal: gimbal measurement audit (objective=measured_sg)

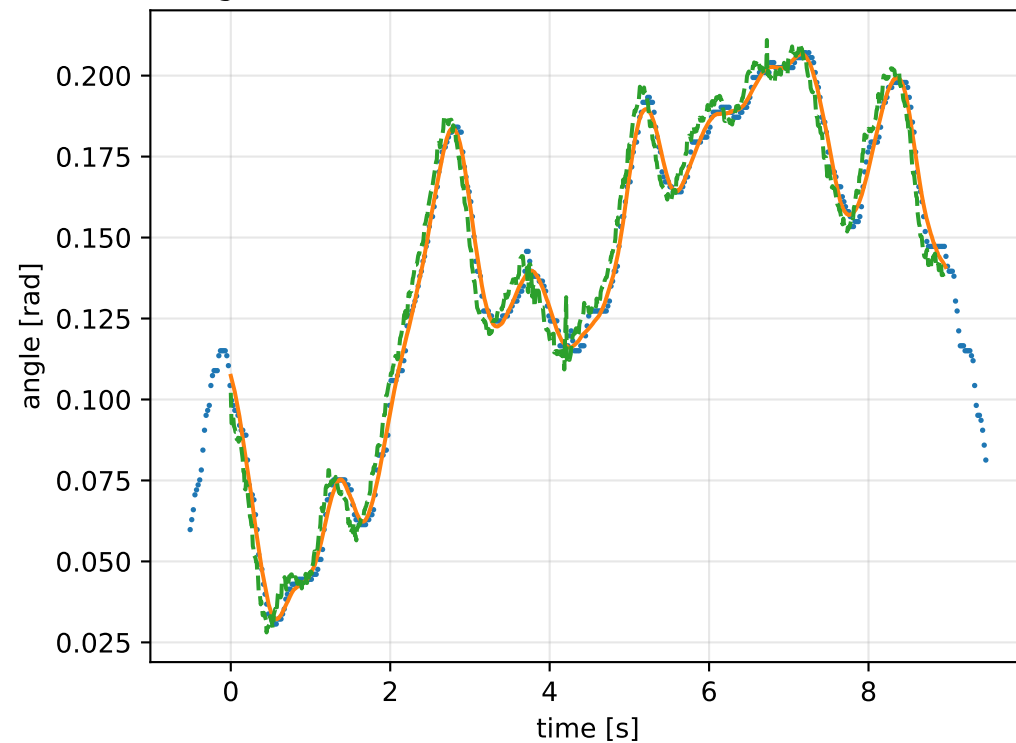
gimbal 1: RMSE=0.006345, max=0.01494 rad



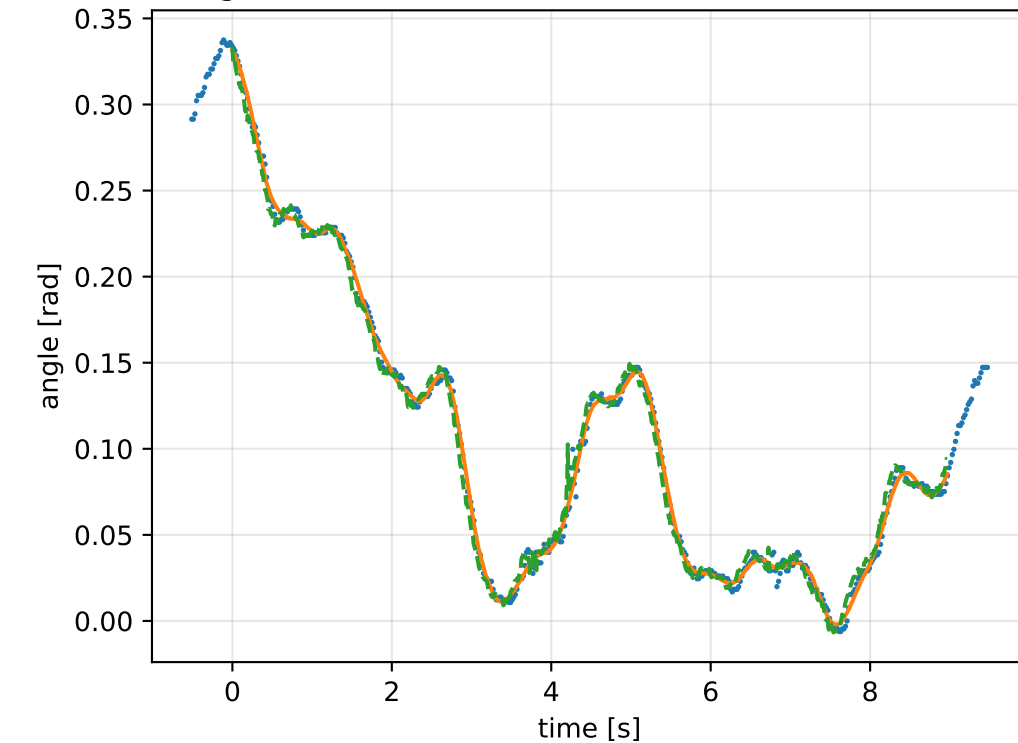
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

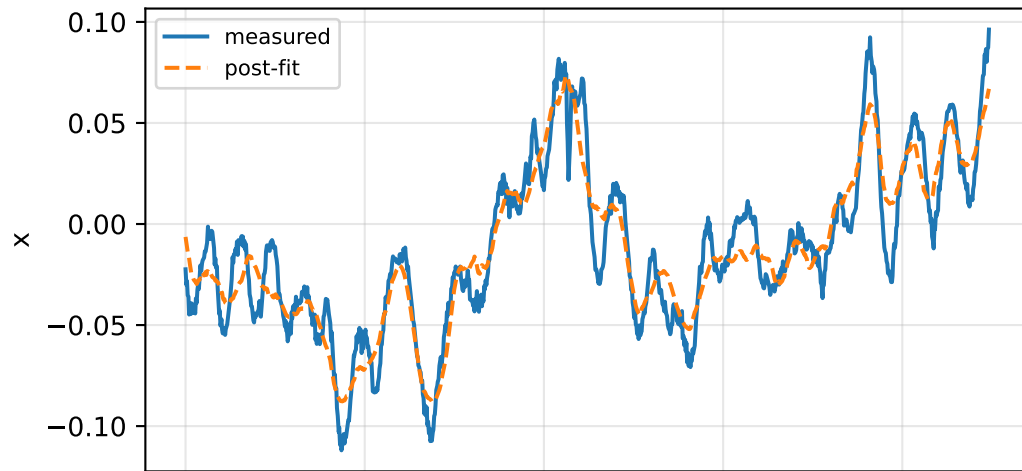


gimbal 4: RMSE=0.006321, max=0.03642 rad

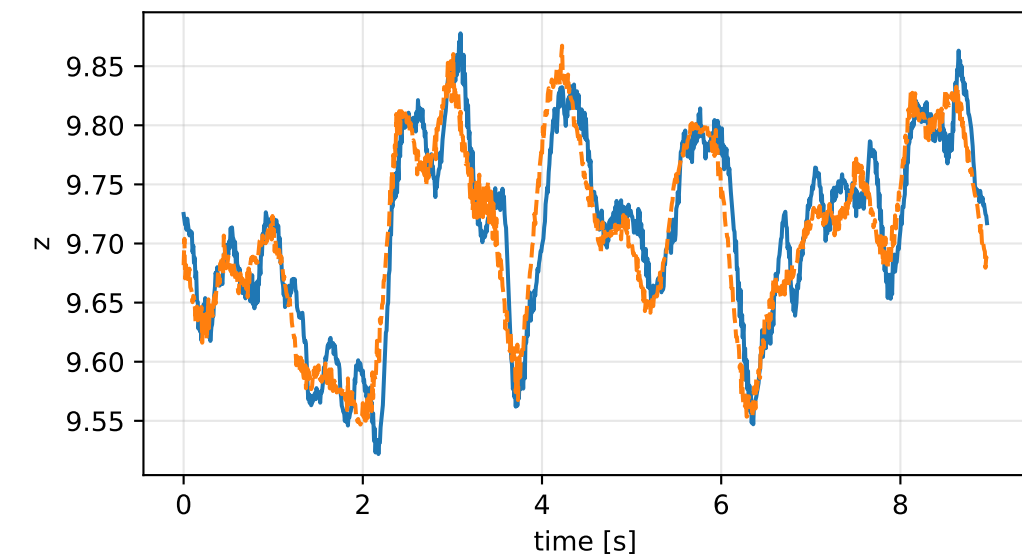
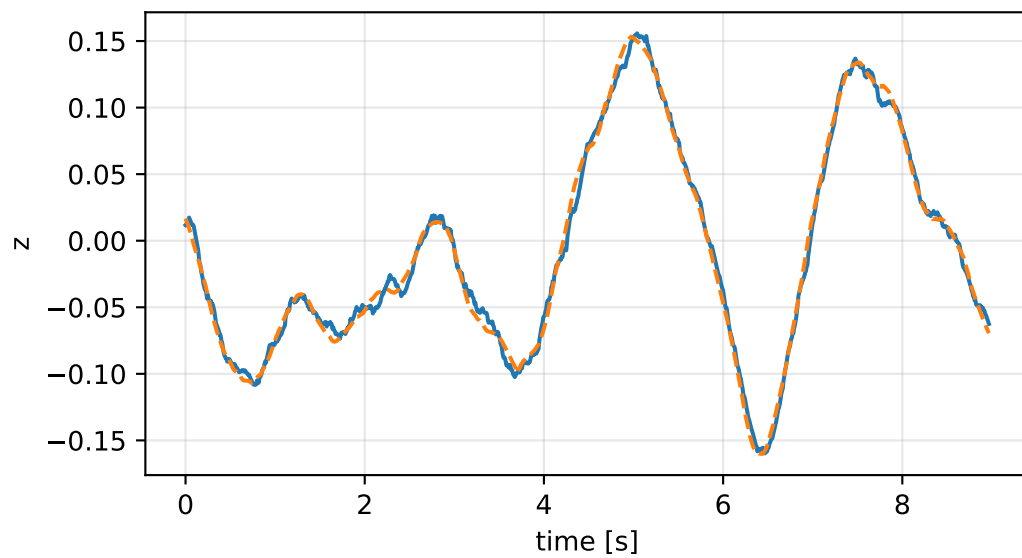
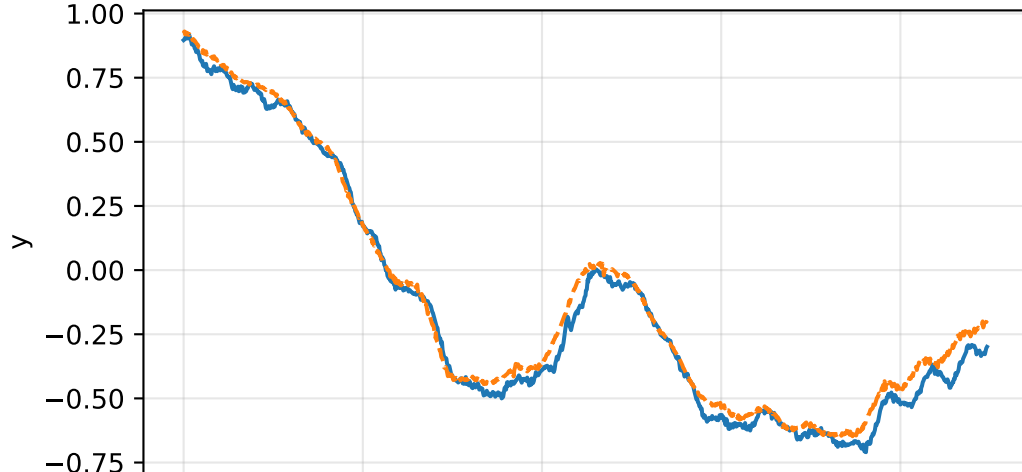
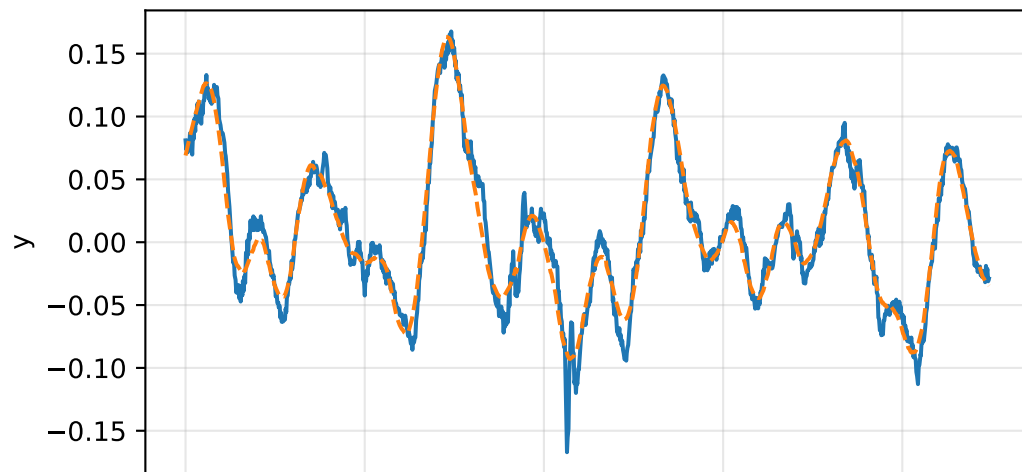
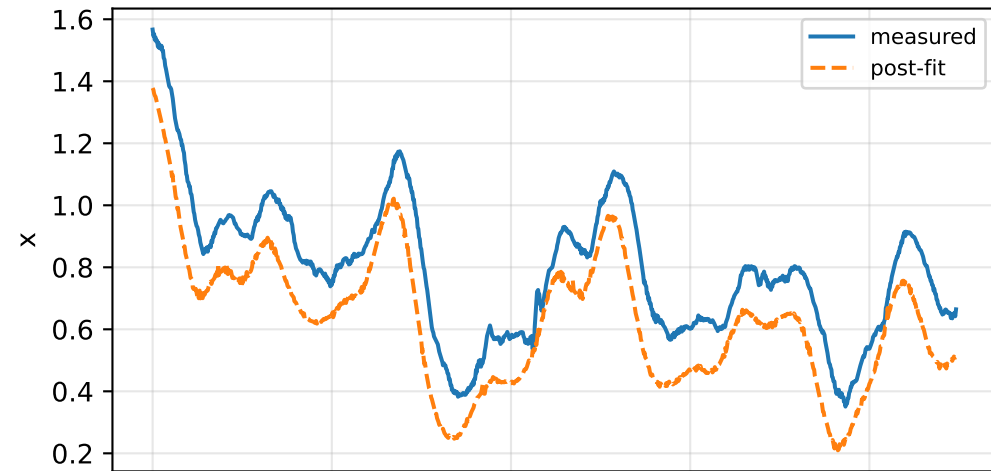


inertia_over_mass_xy_nominal: IMU comparison (diagnostic only)

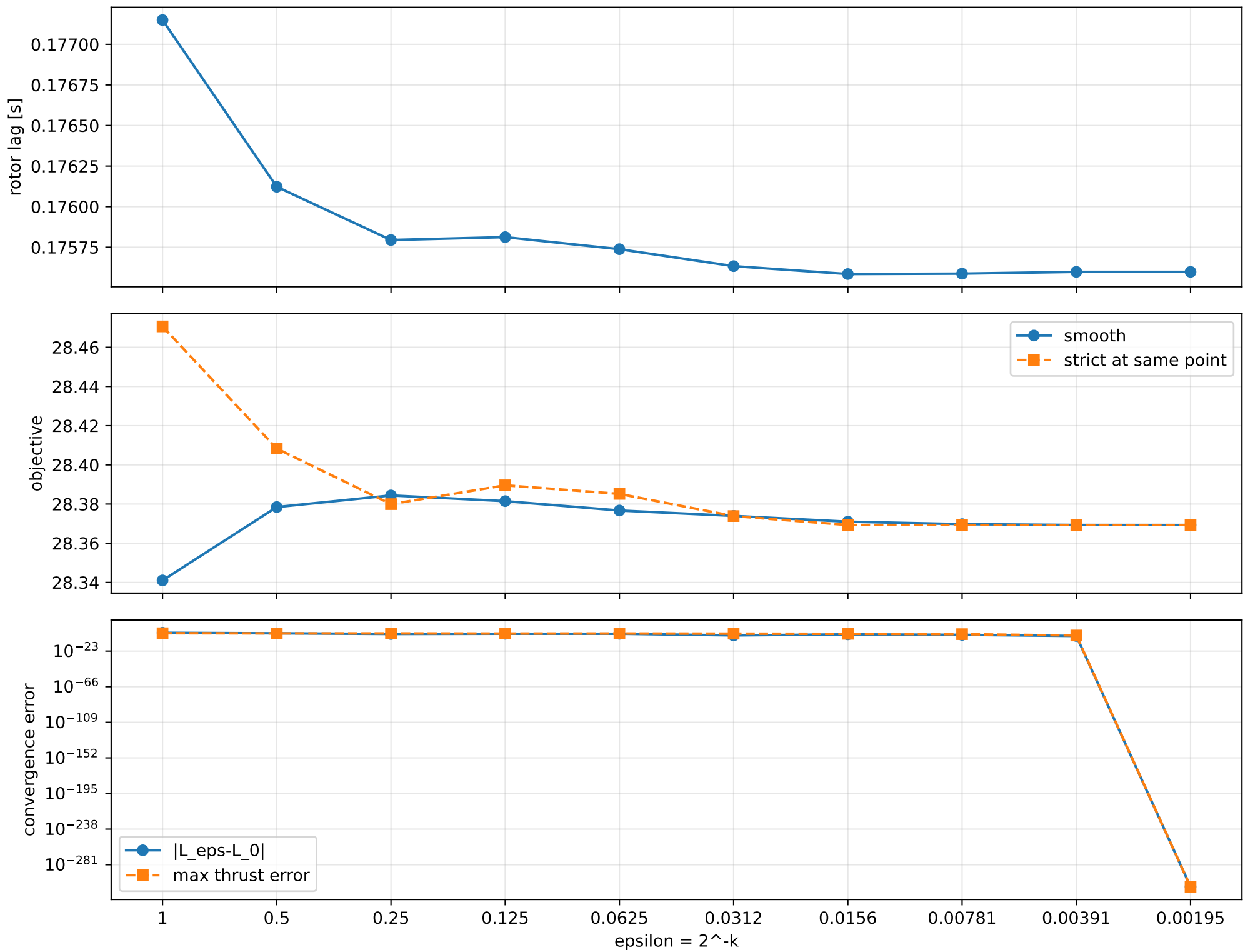
gyro [rad/s]



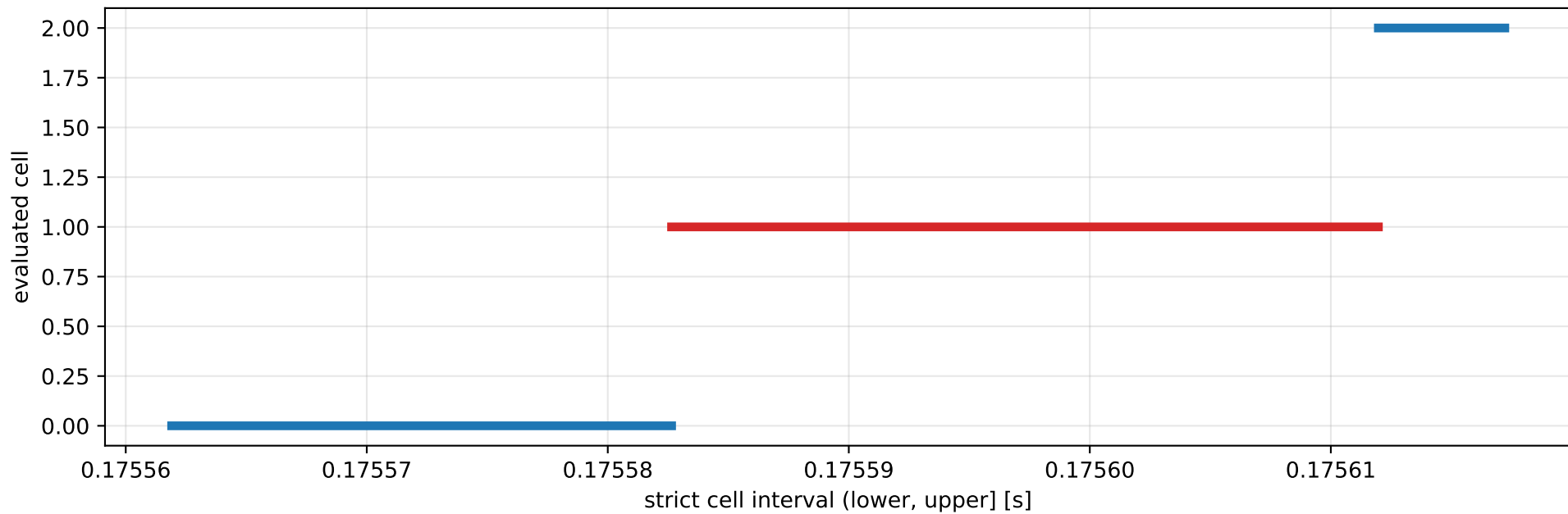
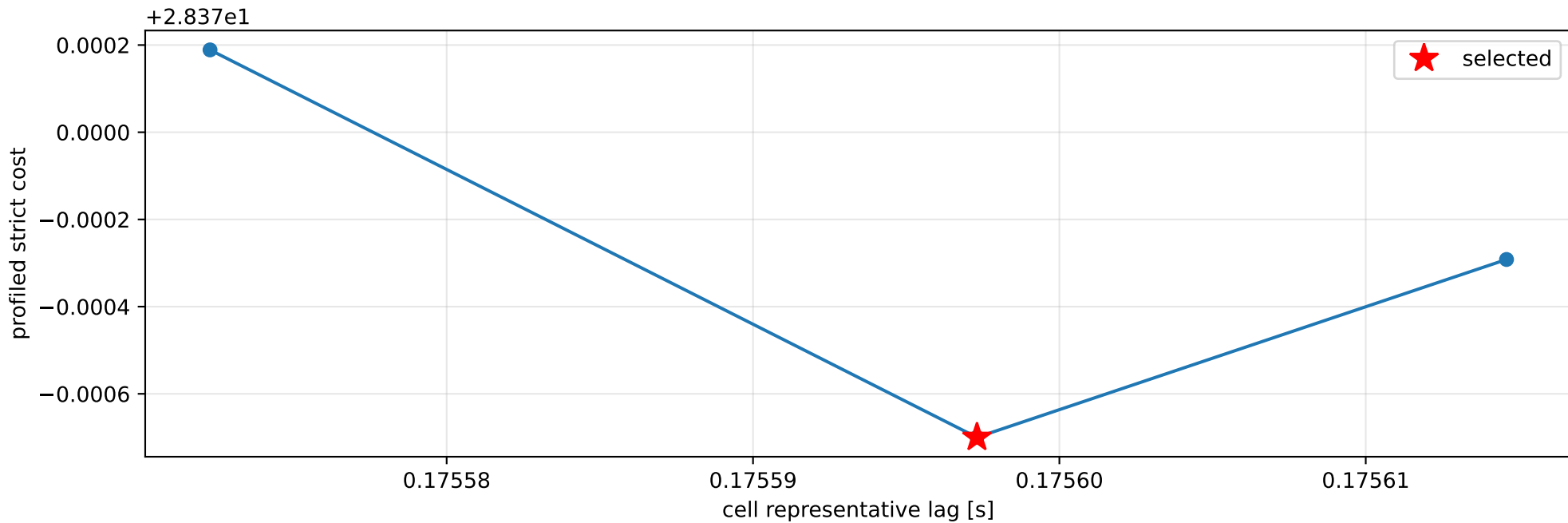
specific force [m/s^2]



inertia_over_mass_xy_nominal: smooth-to-strict continuation



inertia_over_mass_xy_nominal: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175598075 s; data boundary=False

inertia_over_mass_xy_nominal: parameter estimate

Case: inertia_over_mass_xy_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.95024909

inertia [kg m²]:

[[1.15781221e+00 -1.53232217e-06 8.27196590e-02]

[-1.53232217e-06 5.44013536e-01 -1.52382147e-02]

[8.27196590e-02 -1.52382147e-02 6.39583873e-01]]

CoG body [m]: [-0.01956324 -0.05535955 -0.00471323]

force effectiveness: [2.19849885 2.15934431 1.13407385 1.37814529]

scale-free J/m [m²]:

[[2.33889687e-01 -3.09544458e-07 1.67102013e-02]

[-3.09544458e-07 1.09896194e-01 -3.07827232e-03]

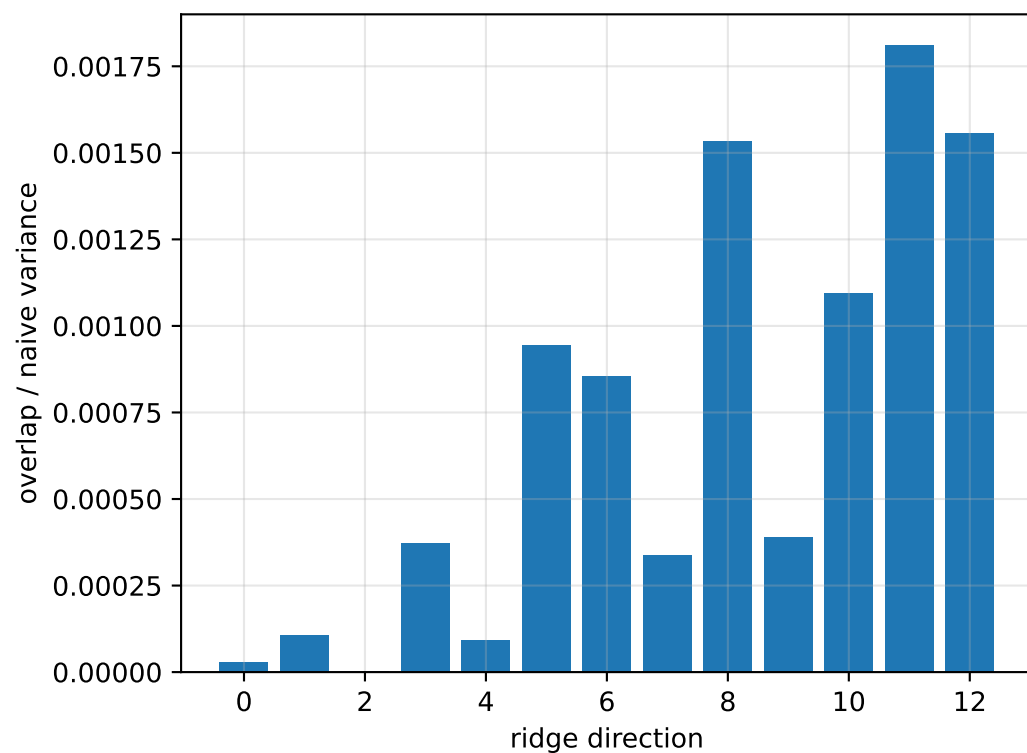
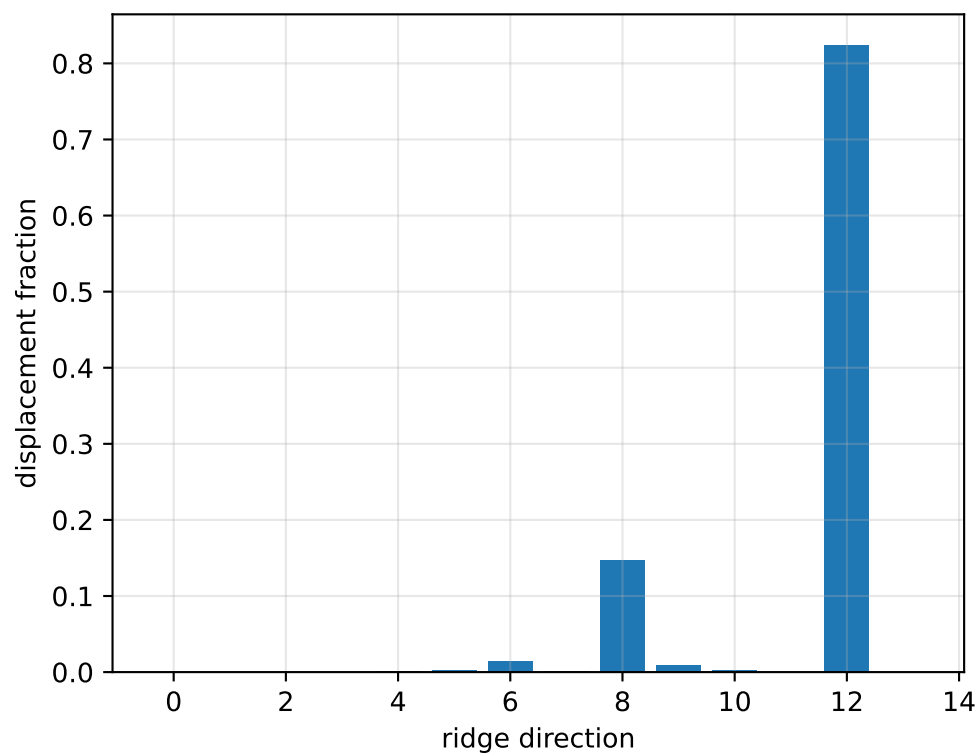
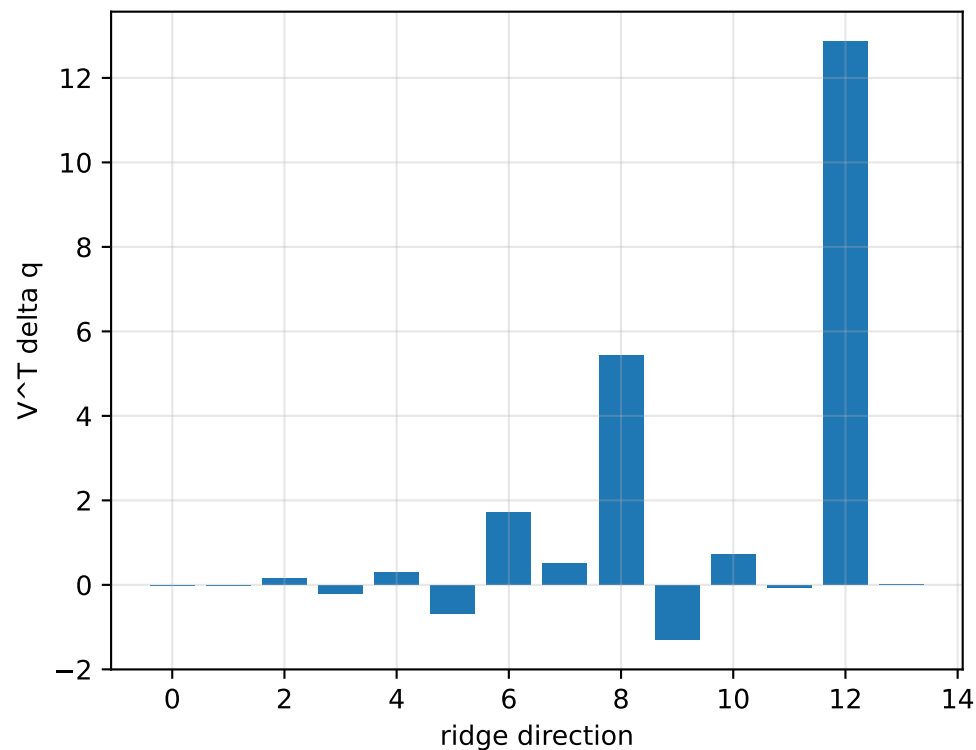
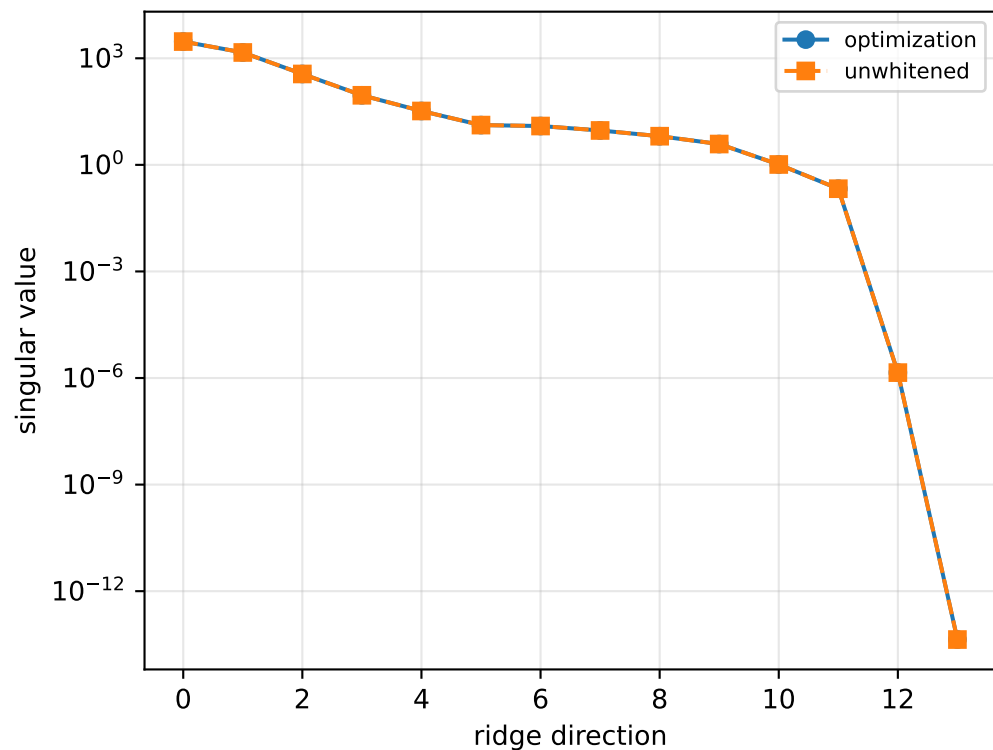
[1.67102013e-02 -3.07827232e-03 1.29202362e-01]]

scale-free f/m: [0.44411883 0.43620922 0.2290943 0.27839918]

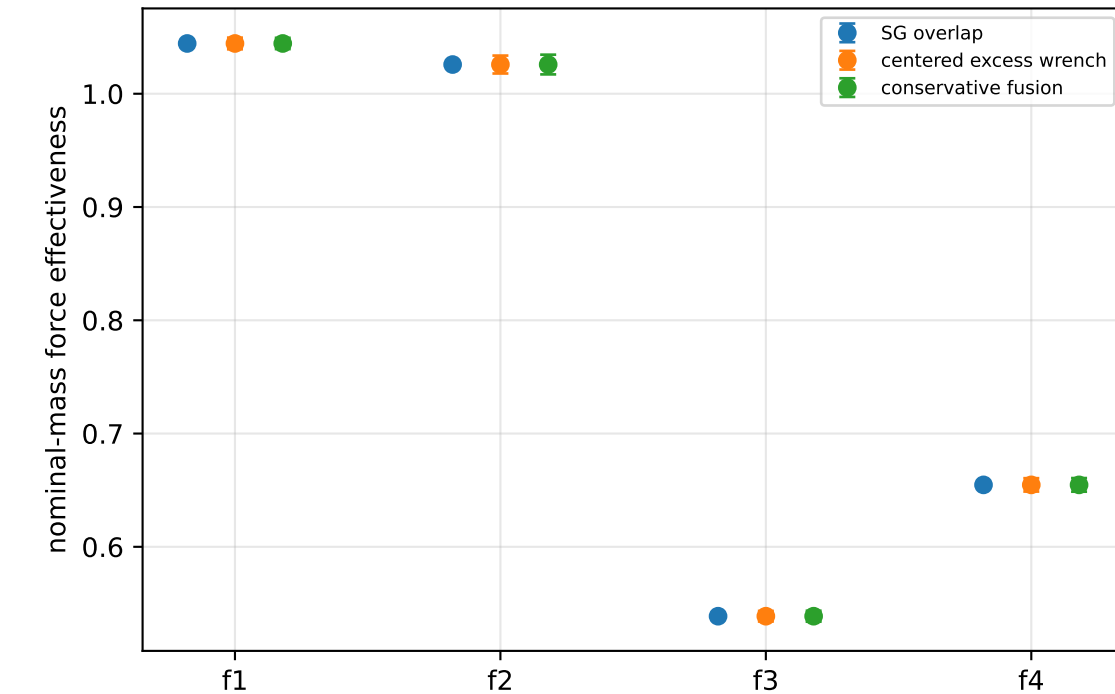
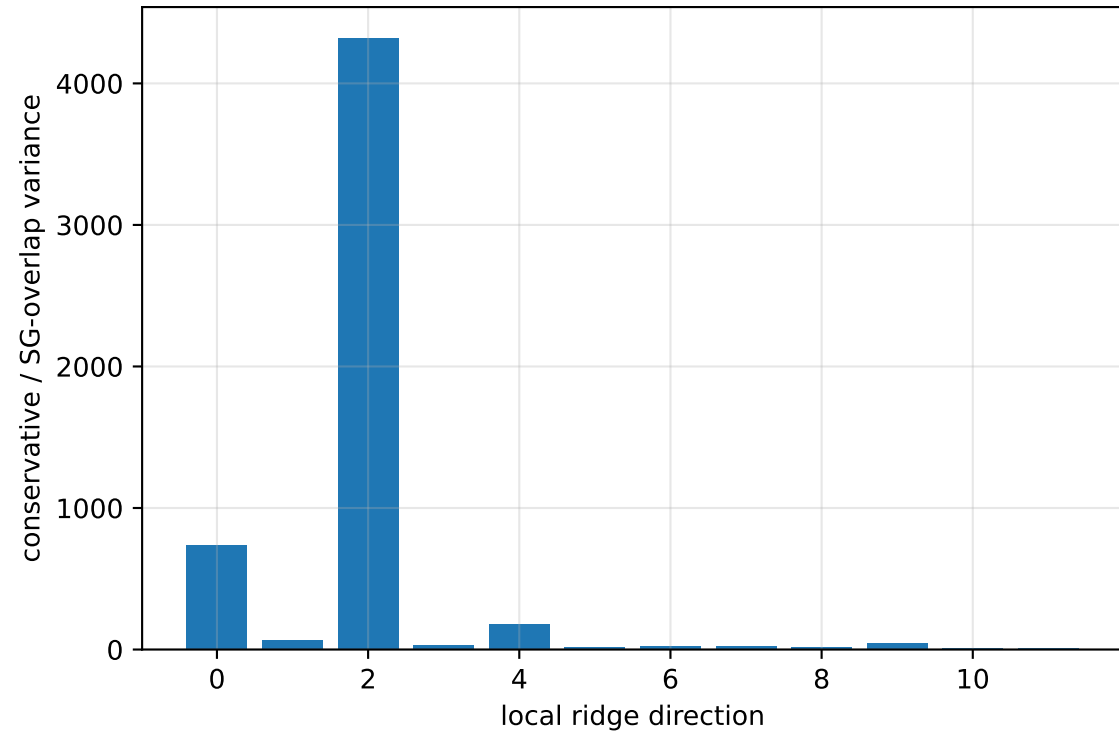
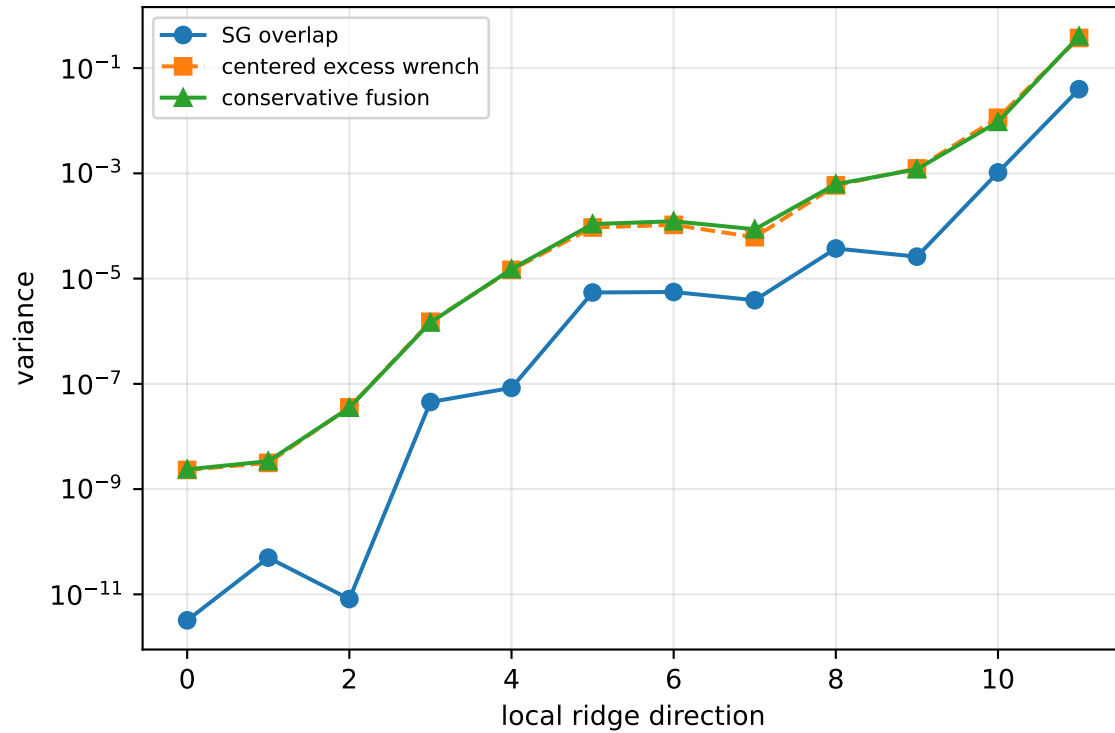
rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

inertia_over_mass_xy_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=7.58e-14)



inertia_over_mass_xy_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 415.7$
wrench-to-acceleration recovery max error = $2.97\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

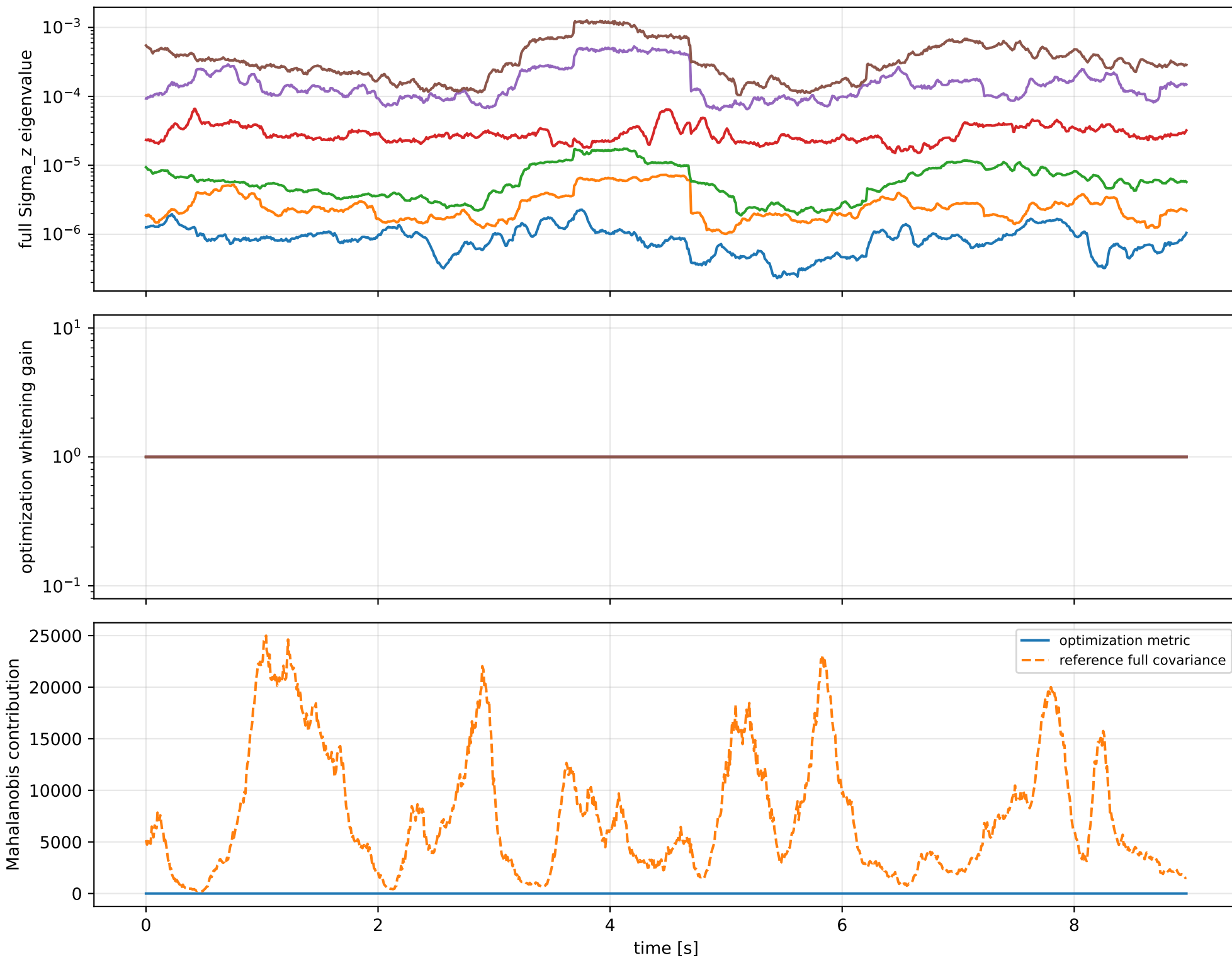
index 11: variance=0.4028, ratio=10.12
 $\log_{\text{mass}}=+0.00167$; second_moment_diag_1=+0.00714
second_moment_diag_2=-0.00305; second_moment_diag_3=-0.0129
second_moment_offdiag_12=+0.995; second_moment_offdiag_13=-0.0967
second_moment_offdiag_23=+0.0185; cog_x=-0.00066
cog_y=+0.000299; cog_z=-0.000521
log_force_eff_1=+0.00371; log_force_eff_2=-0.00286
log_force_eff_3=+0.00129; log_force_eff_4=+0.00501

index 10: variance=0.00952, ratio=9.134
 $\log_{\text{mass}}=-0.0194$; second_moment_diag_1=-0.0178
second_moment_diag_2=-0.0345; second_moment_diag_3=+0.139
second_moment_offdiag_12=+0.0978; second_moment_offdiag_13=+0.982
second_moment_offdiag_23=-0.00262; cog_x=+0.00624
cog_y=-0.000826; cog_z=-0.000629
log_force_eff_1=-0.0369; log_force_eff_2=+0.00625
log_force_eff_3=+0.0219; log_force_eff_4=-0.058

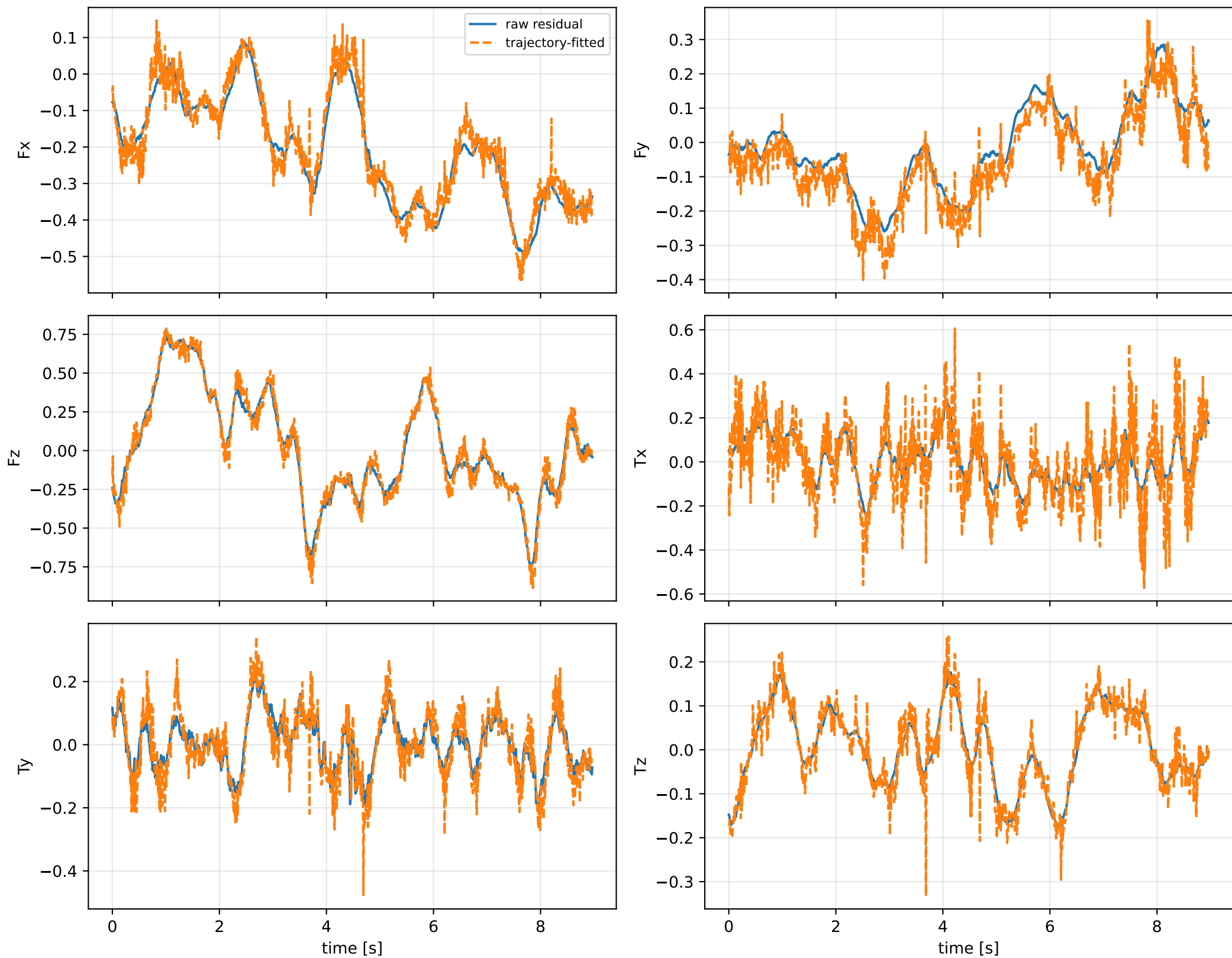
index 9: variance=0.001196, ratio=45.75
 $\log_{\text{mass}}=+0.168$; second_moment_diag_1=+0.000465
second_moment_diag_2=-0.895; second_moment_diag_3=+0.0406
second_moment_offdiag_12=-0.00205; second_moment_offdiag_13=-0.0261
second_moment_offdiag_23=-0.217; cog_x=+0.00479
cog_y=+0.000379; cog_z=+0.0105
log_force_eff_1=+0.161; log_force_eff_2=+0.173
log_force_eff_3=+0.215; log_force_eff_4=+0.136

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_xy_nominal: optimization vs reference covariance



inertia_over_mass_xy_nominal: wrench / closure (force max 1.47e-14, torque max 1.11e-16)



inertia_over_mass_xy_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_xy_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: 0e9e1cf84b655f89c02f04503cdbe3c8cb0feb8b35995087c0bc407814199394

data objective: 28.3693001498

prior objective: 5.53805595917e-11

total objective: 28.3693001499

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_xy_nominal (inertia_over_mass_m2)

components: xy

target: [-3.09533934e-07]

std: [1.e-06]

final: [-3.09544458e-07]

error: [-1.05243109e-11]

standardized residual: [-1.05243109e-05]

factor objective: 5.53805595917e-11